

HASSAN MAHMOOD

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EDUCATION

Northeastern University

Boston, MA

Ph.D. Computer Science (Master's Degree Awarded in 2024)

Jan. 2027 (Expected)

Advisor: Dr. Ehsan Elhamifar

Research Interests: Generative Models, Efficient Adversarial Attacks and Robustness, Computer Vision

National University of Sciences and Technology

Islamabad, Pakistan

Bachelor of Software Engineering

July 2018

Advisor: Dr. Faisal Shafait

Research Interests: Computer Vision, Real-time Vehicle Detection and Localization

PUBLICATIONS

- **Hassan Mahmood**, Alina Oprea, Ehsan Elhamifar, "Clean-Label Backdoor Attacks for Diffusion Models", (Submission to NeurIPS 2026)
- **Hassan Mahmood**, Ehsan Elhamifar, "Compositional Targeted Multi-Label Universal Perturbations", CVPR 2025. [PDF]
Spotlight presentation — *Uncertainty Quantification for Computer Vision Workshop* (CVPR 2025).
- **Hassan Mahmood**, Ehsan Elhamifar, "Semantic-Aware Multi-Label Adversarial Attacks", CVPR 2024. [PDF]
- S. R. Qasim, **Hassan Mahmood**, Faisal Shafait, "Rethinking Table Recognition using Graph Neural Networks", ICDAR 2019. [PDF]

WORK EXPERIENCE

PhD Intern

May 2026 – Aug. 2026

Google Cloud AI

Seattle, WA

Hierarchical Multi-Agent System for Fleet-Scale Cloud Root-Cause Analysis

- **Problem:** Existing agentic triage took **16 hours per 10K machines** and required engineer-initiated, engineer-supervised runs — intractable across a fleet of millions of VMs — while saturating context on multi-log investigations, yielding attention dilution, circular reasoning, and inconclusive root cause.
- **Solution:** Built the specialized-agent layer of a **hierarchical orchestrator** → **workflow** → **component-agent** system; the router decomposes each investigation and dynamically dispatches agents scoped to a single log domain with minimal toolsets (BigQuery telemetry, log search, host probes), each returning compressed evidence rather than raw traces.
- Isolated per-agent context with parallel retry and independent verification turns one-shot investigation into **pass@k**, enabling **fully automated fleet-wide straggler analysis** in place of manually driven per-batch runs. Cut wall-clock from **16h to ~2h per 10K machines** and inconclusive root-cause triage from **20% to ~8%**; deployed to SRE users.

Graduate Research Assistant

Aug. 2020 – Present

NOVA Lab, Northeastern University

Boston, MA

Project: Multi-Task Video Understanding with Vision-Language Models (ongoing work)

- **Problem:** Limited VLM context windows force sparse frame selection on long **multi-task** egocentric videos, and uniform sampling often misses interleaved actions across tasks.
- Developing methods to allocate scarce VLM capacity via action-coverage-aware **frame selection** and selective **token merging**, enabling robust zero-shot **long-form temporal understanding** under tight compute constraints.

Project: Backdoor Attacks on Diffusion Generative Models

- **Problem:** Stealthy poisoning of a small fraction of diffusion-model training data is challenging; naive small-perturbation based backdoor attacks generate low-density poisoned samples which the model's denoising process treats as noise and effectively ignores, limiting poisoning effectiveness.
- Developed a **meta-learning** framework that optimizes the adversarial input modifications that allow the standard diffusion training to learn better score estimates on the poisoned samples - achieving ~ 20% better poisoning success rate.

Project: Model Responsibility Abdication Auditing

- Built a multi-agent simulation using Anthropic's Petri framework to evaluate **responsibility abdication**, where autonomous LLM employees over-trust auditor tools under time pressure and misalign by giving up their own judgment and accountability.

Project: Text-based Error Generation using LLM Prompt Tuning

- **Problem:** Generating realistic, valid text-based error descriptions for procedural task videos is challenging; standard Llama-2 prompting often yields invalid, low-quality, or uncontrollable errors.
- Developed a **prompt-tuning** framework for **Llama-2** that uses adversarial discrete optimization and iterative feedback from a secondary Llama-2 verifier to improve realistic error descriptions, achieving $\sim 35\%$ higher validity than the baseline.

Project: Universal Multi-Label Adversarial Attacks

- **Problem:** Generating targeted universal adversarial perturbations in multi-label models requires handling every label combination, causing exponential complexity.
- Developed a compositional optimization framework assuming label-independence to generate scalable **targeted universal perturbations** for multi-label models, reducing computational complexity from **exponential** to **linear**. Empirically demonstrated approximate linearity of universal adversarial regions in deep models.

Project: Multi-Label Adversarial Attacks

- **Problem:** Existing multi-label attacks fail on large datasets and often modify the prediction of correlated non-target labels.
- Identified **negative gradient correlation** as the cause of suboptimal multi-label adversarial attacks. Developed a **knowledge graph-based** framework that generates semantic-aware attacks and addresses the negative correlation, enabling efficient attacks with $\sim 20\%$ higher success and reduced runtime on large datasets (PASCAL-VOC, NUS-WIDE, OpenImages).

Graduate Research Assistant

Aug. 2019 – July 2020

CBL Lab, University of Houston

Houston, TX

Project: Explainable and Interpretable Deep Learning in Healthcare

- Built deep learning models for cardiovascular disease prediction on the MESA dataset, achieving a 6% increase in F1-score compared to prior approaches. Additionally, developed interpretability analyses using SHAP and LIME to provide insights.
- Designed a privacy-preserving affective computing framework that leveraged temporal gait modeling to analyze customer behavior, directly contributing to the acquisition of a **\$100K** research grant.

AI Researcher

July 2018 – Aug. 2019

Deep Learning Lab, National Center of Artificial Intelligence

Islamabad, Pakistan

Project: Table Structure Recognition using Graph Neural Networks

- **Problem:** Accurately recognizing table structures in document images is challenging due to the varied and complex layouts frequently encountered in real-world data; existing methods often struggle to generalize.
- Developed a novel **Graph Neural Network** (GNN)-based approach for recognizing table structures in document images, effectively capturing relationships between table cells and structural elements. This method robustly handles complex and irregular table layouts, improving accuracy by **37%** over baseline methods.
- Created and open-sourced a synthetic dataset generation algorithm for tabular document images ([[GitHub](#)]).

Other Machine Learning Projects

- Designed an outfit classification and recommendation system for US-based startup **Trendage**.
- Authored an e-book tutorial on machine learning and deep learning for **AI Sciences**.
- Implemented a deep learning character recognition pipeline for LCD queue displays (grant proposal, Talents Fellowship Program, Machine Learning Lab, University of Trieste).

ENGINEERING SKILLS

- **Programming Languages:** Python, C/C++, Java, SQL, Javascript, Matlab
- **ML Frameworks/Tools:** Pytorch, Keras, Tensorflow, Jupyter, OpenCV, Numpy, Scipy
- **Others:** Git, AWS, Selenium, Docker, Flask, Wireshark, TCP/IP Protocol, DNS Protocol

SERVICES

- Served as a reviewer for TPAMI journal, CVPR, AAAI, ICCV, and ICLR conferences.
- Supervised an undergraduate research intern for six months at University of Houston.

AWARDS

- Selected as one of the 190 students from a pool of 2500+ candidates across Pakistan for fully funded UGRAD semester exchange program sponsored by the U.S. Department of State (worth $\approx$ **\$22,500**).
- Prime Minister National ICT R&D Merit Scholarship for fully funded undergraduate degree and monthly stipend for four years (worth $\approx$ **\$12,000**).
- Bronze medal and merit scholarship for 3rd standing (in a total of $\approx$ **40,000** students) in computer science higher education exams in Pakistan - FBISE, Islamabad.